Project: Worked examples
Title: Fatigue damage (Palmgren–Miner)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Cumulative fatigue damage of a stress-range spectrum against an S–N curve of one slope through a reference point, with an optional endurance limit below which a range does no damage; the spectrum repetitions to failure.
N_i = N_ref·(S_ref/S_i)^m, unbounded when S_i ≤ S_e. d_i = n_i/N_i. D = Σ d_i; repetitions to failure = 1/D. The damage passes when D ≤ 1.
Ref: BS EN 1993-1-9 §8 — Fatigue resistance (S–N curves by detail category)
Ref: Shigley's Mechanical Engineering Design §6-15 — Cumulative fatigue damage

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
S_ref	Reference stress range (detail category or material fatigue strength)	71	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
N_ref	Cycles at the reference stress range	2000000		spec: SPEC-MECH-03 design criteria	10000 … 1000000000
m	S–N slope	3		spec: SPEC-MECH-03 design criteria	2 … 15
blocks	Stress-range blocks	3 row(s)		spec: LOAD-SCHEDULE-01	any
blocks[1]		S = 100 MPa; n = 100000			
blocks[2]		S = 60 MPa; n = 500000			
blocks[3]		S = 40 MPa; n = 2000000			

3. Assumptions

A1. Damage accumulates linearly and independently of the order of the blocks. (Palmgren–Miner rule.)
A2. One S–N slope over the whole range; no second slope below a knee. (Simplified curve; a two-slope curve is entered as two sheets or a lower slope.)

4. Working

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[4] N_1 = N_ref · (S_ref / S_1)^m
= 2000000 × (71 MPa / 100 MPa)^3
= 715820
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BS EN 1993-1-9 §8

Life at block 1

[5] d_1 = n_1 / N_1
= 100000 / 715820
= 0.1397
Damage of block 1
[6] N_2 = N_ref · (S_ref / S_2)^m BS EN 1993-1-9 §8
= 2000000 × (71 MPa / 60 MPa)^3
= 3314000
Life at block 2
[7] d_2 = n_2 / N_2
= 500000 / 3314000
= 0.15088
Damage of block 2
[8] N_3 = N_ref · (S_ref / S_3)^m BS EN 1993-1-9 §8
= 2000000 × (71 MPa / 40 MPa)^3
= 11185000
Life at block 3
[9] d_3 = n_3 / N_3
= 2000000 / 11185000
= 0.17882
Damage of block 3
[10] D = ∑ d_i Shigley §6-15
= 0.1397 + 0.15088 + 0.17882
= 0.46939
Total damage (Palmgren–Miner)
[11] R_f = 1 / D
= 1 / 0.46939
= 2.1304
Spectrum repetitions to failure

5. Results

SYMBOL	VALUE	DESCRIPTION
D	0.46939	Total damage
R_f	2.1304	Repetitions to failure
N_1	715820	Life at block 1
d_1	0.1397	Damage of block 1
N_2	3314000	Life at block 2
d_2	0.15088	Damage of block 2
N_3	11185000	Life at block 3
d_3	0.17882	Damage of block 3

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Cumulative damage D ≤ 1	D = 0.46939	D ≤ 1	0.469	Shigley §6-15

7. Applicability

No warnings.

8. Verification

Template mech.fatigue-miner@1.0.0 Content hash 9a0ade9f4660b0be Engine calc-engine/1.0.0
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